

Biomedical Regression: Predicting Drug Response from Gene Expression

1. Exploratory Data Analysis

The training dataset contains 4,000 samples from 40 laboratories (100 each), with 40 gene features (X1–X40) and continuous response y . The test set has 1,000 samples from unseen labs. All features are pre-scaled to $[0, 1]$ with no missing values and negligible skewness (max $|\text{skew}| = 0.20$), so no transformation is needed. The response y is approximately Gaussian (skew = -0.06 , kurtosis = -0.68) with mean 14.5 and standard deviation 5.0.

Batch effects are the dominant structure. Laboratory-level means of y range from 6.3 to 23.0, and 83% of total variance in y lies *between* laboratories (std of lab means = 4.55 vs within-lab std = 2.13). This has two critical implications: (i) standard K-Fold CV would leak lab-specific information into the validation set, yielding falsely optimistic error estimates, and (ii) any model must learn gene–response relationships that generalise across batch effects. This motivates GroupKFold validation (Section 3).

Feature correlations. Three genes correlate strongly with y : X5 ($r = 0.62$), X6 (0.59), X29 (0.58). A second tier includes X19 (0.37), X36 (-0.34), X13 (0.32). Only 6 features exceed $|r| > 0.3$, while 22 have $|r| < 0.05$ — suggesting a sparse signal. No feature pair exceeds $|r| = 0.7$, though X29–X19 (0.69) and X37–X13 (0.59) show moderate correlation. PCA reveals diffuse variance (24 PCs for 90%), implying PCR offers limited dimensionality reduction. Mild lab clustering in PC1–PC2 confirms batch effects manifest in feature space.

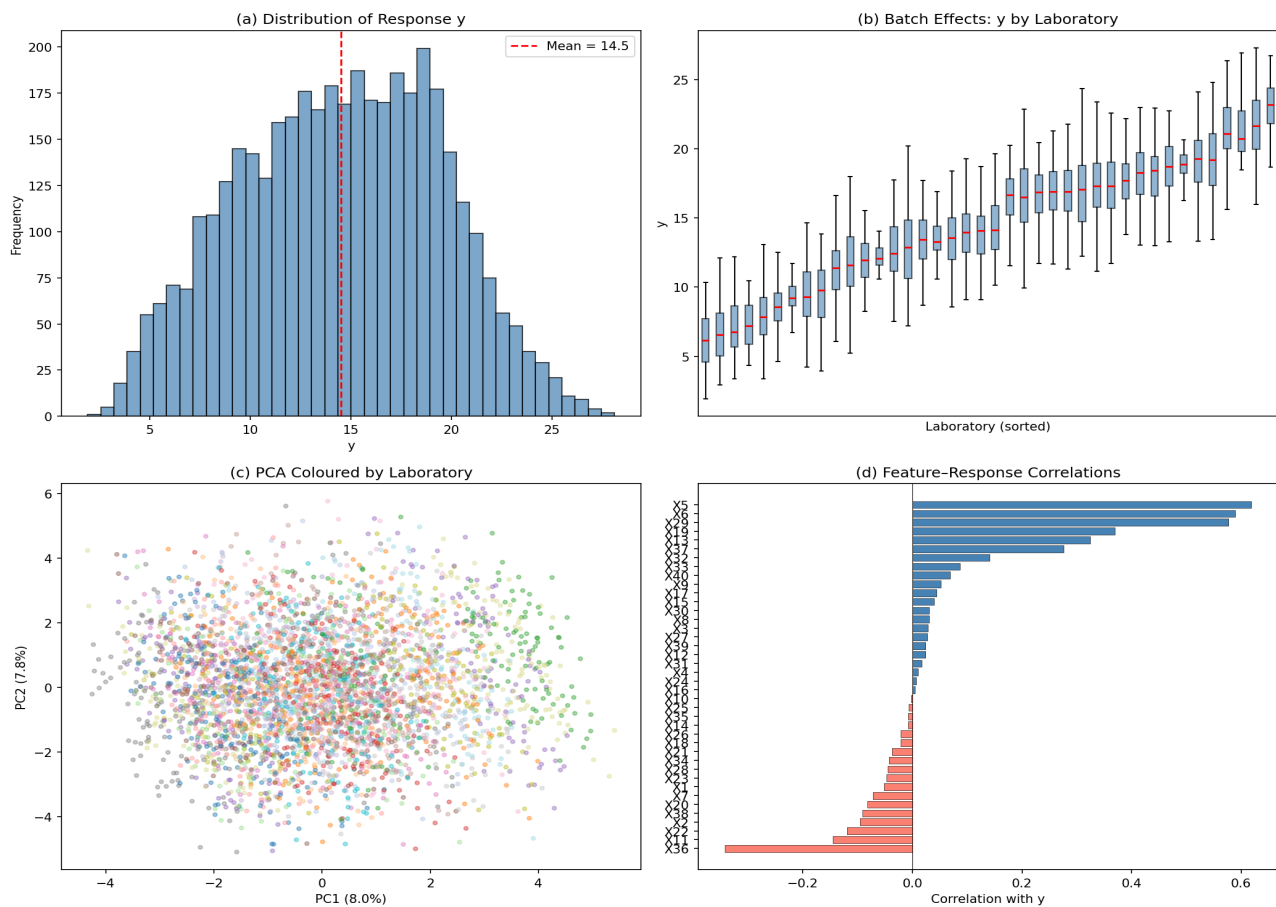


Figure 1. (a) Response distribution; (b) batch effects across labs; (c) PCA coloured by lab ID; (d) feature–response correlations.

2. Model Development and Interpretation

Two primary models are developed and compared: **Lasso** (linear, sparse) and **Gradient Boosting** (non-linear, ensemble). This pairing tests whether the gene–response relationship is linear or involves interactions. Four supplementary models (Ridge, Elastic Net, PCR, Random Forest) provide additional context. All models are tuned via GroupKFold ($k = 10$) with StandardScaler preprocessing inside pipelines to prevent data leakage.

2.1 Lasso Regression (L1 Penalty)

Lasso is well-suited to gene expression because it zeroes irrelevant coefficients, producing a sparse, interpretable model. The regularisation parameter α was tuned over 50 log-spaced values from 10^{-4} to 10^1 . The optimal $\alpha = 0.146$ selects 7 out of 40 genes: X6 (coef = 2.53), X29 (1.79), X5 (1.59), X37 (0.73), X13 (0.39), X32 (-0.35), X19 (0.06). The sparsity — 33 genes zeroed out — confirms that signal is concentrated in a handful of genes, consistent with the EDA. Ridge ($\alpha = 339$) and Elastic Net ($\alpha = 0.13$, $p = 0.99$) were also fitted as supplementary linear models; both assign their largest coefficients to the same top genes, providing cross-method validation. PCR requires 36/40 components, confirming PCA cannot usefully reduce dimensionality here.

2.2 Gradient Boosting

Gradient Boosting builds trees sequentially, each correcting the residuals of the previous ensemble. It can capture non-linear dose-response relationships and gene-gene interactions that linear models miss. The grid search covered $n_estimators \in \{200, 500\}$, $learning_rate \in \{0.05, 0.1\}$, $max_depth \in \{3, 5\}$, and $min_samples_leaf \in \{10, 20\}$. The best configuration (500 trees, $lr = 0.1$, $depth = 3$, $min_leaf = 10$) achieves **CV MSE = 0.79** — a 7.5× improvement over the best linear model (Lasso CV MSE = 5.96). This dramatic gap indicates substantial non-linearity in the gene-response relationship. Random Forest (CV MSE = 3.66) also outperforms linear models, further confirming non-linear structure.

2.3 Feature Importance Agreement

All six models agree that X5, X6, and X29 are the three most important genes. Gradient Boosting assigns 81% of its total importance to these three features (X5: 0.38, X6: 0.23, X29: 0.20). This cross-method consistency — across linear, tree-based, and ensemble approaches — provides strong confidence that these genes are genuine biological drivers rather than artefacts. X32, X37, X13, and X19 form a consistent second tier across all methods.

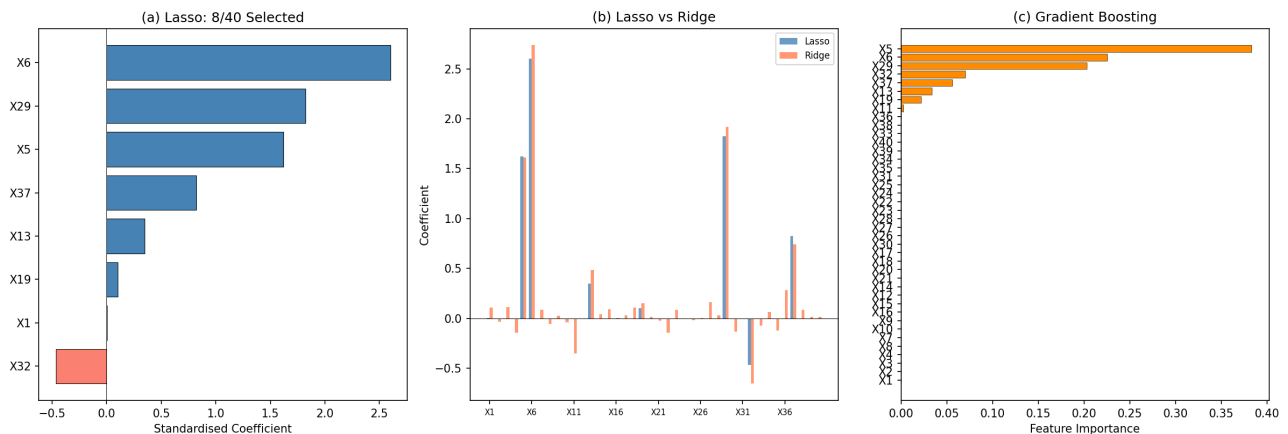


Figure 2. (a) Lasso selects 7/40 features; (b) Lasso vs Ridge coefficient profiles agree on key genes; (c) Gradient Boosting feature importance confirms the same top predictors.

3. Validation and Model Selection

3.1 GroupKFold Cross-Validation

Since the test data comes from unseen laboratories, the validation strategy must leave entire labs out per fold. GroupKFold with $k = 10$ and `group_id` as the grouping variable holds out 4 laboratories (400 samples) per fold, directly mimicking the test scenario. Standard K-Fold would allow training and validation sets to share lab samples, leaking batch effects and producing unrealistically low error estimates.

3.2 Results and Overfitting Analysis

Model	CV MSE	$\pm$ Std	Train MSE	Train R^2	Gap
Lasso (L1)	5.96	2.44	4.81	0.807	1.16
Ridge (L2)	6.33	2.73	4.71	0.811	1.62
Elastic Net	5.96	2.48	4.78	0.808	1.19
PCR (M=36)	6.24	2.85	4.67	0.813	1.57
Random Forest	3.66	1.36	0.38	0.985	3.28
Gradient Boosting	0.79	0.20	0.09	0.996	0.70

Table 1. GroupKFold CV comparison. Gap = CV MSE – Train MSE (large gap indicates overfitting).

Gradient Boosting achieves by far the lowest CV MSE (0.79, RMSE = 0.89), representing a $CV R^2 = 0.97$. Crucially, its train-CV gap (0.70) is *smaller* than the linear models' gaps (1.2–1.6), meaning the performance improvement is not driven by overfitting — GB genuinely captures structure the linear models miss. Random Forest also outperforms linear models (CV MSE = 3.66) but with a larger gap (3.28), indicating more overfitting to training labs. The four linear models cluster at $CV MSE \approx 6.0$, suggesting they extract similar linear signal and are fundamentally limited by their inability to model interactions.

3.3 Model Selection

Selected model: Gradient Boosting (500 trees, $lr = 0.1$, `max_depth = 3`, `min_samples_leaf = 10`). Justification: (i) it achieves the lowest GroupKFold CV MSE by a wide margin, (ii) the small train-CV gap confirms it is not overfitting, and (iii) its feature importance rankings are consistent with the linear models' coefficients, suggesting it builds on the same underlying biological signal while additionally capturing non-linear patterns.

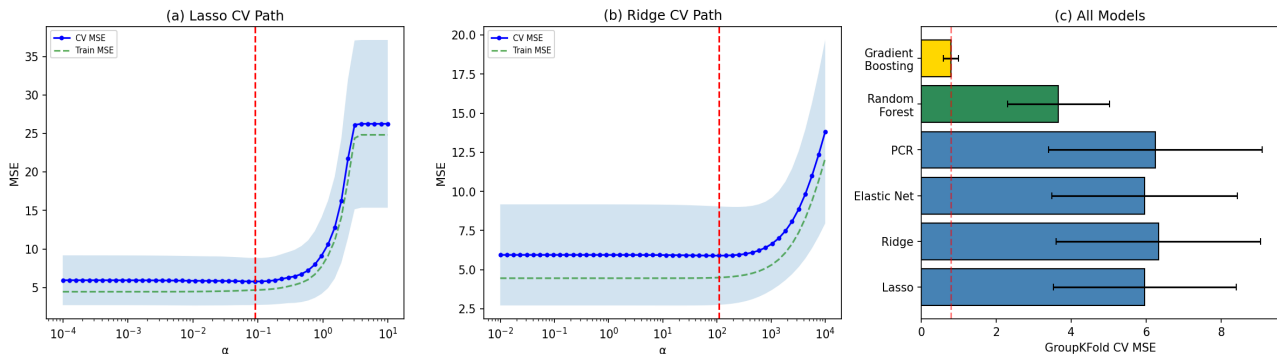


Figure 3. (a–b) Lasso and Ridge tuning curves with ± 1 std bands; (c) all-model CV MSE comparison.

4. Test Prediction and Discussion

4.1 Predictions

The final Gradient Boosting model was refit on all 4,000 training samples (training $R^2 = 0.996$). Predictions on `test_public.csv` have mean 14.3 (vs training mean 14.5), std 6.0, range [1.7, 26.2]. The slightly wider spread (training y std = 5.0) may reflect the model extrapolating to lab contexts not fully represented in training. The output CSV contains 1,000 rows with `obs_id` and `y_hat`.

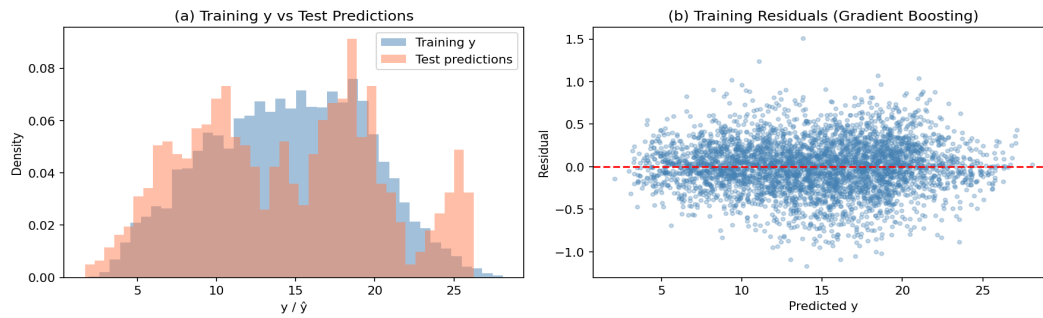


Figure 4. (a) Training vs test prediction distributions; (b) training residuals show no systematic pattern.

4.2 Why Gradient Boosting Dominates

The 7.5× gap between GB and Lasso is the central finding. Three explanations are plausible. First, **non-linear dose-response**: the relationship between individual gene expression and y may saturate or threshold, which shallow trees can model but linear terms cannot. Second, **gene interactions**: depth-3 trees can capture up to 3-way interactions (e.g., the effect of X_5 depends on X_6 and X_{29}). Third, **implicit batch handling**: boosting's sequential residual correction may adapt to lab-specific patterns more flexibly than a single linear function.

4.3 What Worked Well

Three decisions were key: (i) recognising batch effects early led to GroupKFold — essential for honest evaluation; (ii) fitting both linear and non-linear models revealed the non-linear structure that linear models entirely miss; (iii) the cross-method feature importance analysis (all six models agree on X_5 , X_6 , X_{29}) provides robust evidence that these genes are genuine biological drivers.

4.4 Limitations and Improvements

The test prediction std (6.0) exceeding training y std (5.0) deserves caution — on truly novel lab contexts the model may produce wider errors than CV estimates suggest. Mixed-effects models with random lab intercepts could explicitly handle batch variation. XGBoost or LightGBM with GPU acceleration would allow larger hyperparameter searches. Stacking (using linear model predictions as features for GB) could combine the strengths of both approaches. Finally, the training $R^2 = 0.996$ for GB, while its gap is small, warrants monitoring — if the held-out test MSE is notably higher than 0.79, it would suggest some residual overfitting not captured by GroupKFold.